1. If F1: $(y, x) \longmapsto (x,y)$ is bijective then that means that
For every value of $y \in BxA$ there exists an $x \in AxB$ such that $f(x) = y$ meaning it is surjective:
$\forall(x,y) \in AxB\ \exists(y,x) \in BxA\ f(x,y) = (y,x)$ and that
If $x \in AxB$ (a relation between A and B) and $y \in BxA$ (a relation between B and A) then this would mean that there is at most one value of x for every y such that $x = y$ meaning it is injective:
$f(x,y) = f(y,x) \rightarrow (x,y) \in AxB = (y,x) \in BxA$ where $x = y$ and $y = x$
Because F1 is bijective its inverse, $F1^{-1}$, would be
$F1^{-1} = (x,y) \longmapsto (y,x)$

2. F2: $x \longmapsto f^{-1}(g^{-1}(x))$

   Because both f and g are considered bijective it means that for $x \in A$ there exists and

   Let $x \in A$, $y \in B$, $z \in C$ such that $f(z) = y$ and $g(y) = x$ and that $f^{-1}(y) = z$ and $g^{-1}(x) = y$

   This found mean that $f: C \rightarrow B$ and $g: B \rightarrow A$ and thus that $f^{-1}: B \rightarrow C$ and $g^{-1}: A \rightarrow B$

   $f^{-1}(g^{-1}(x)) = f^{-1}(y) = z$ where $z \in C$ which is not equivalent to A as stated in F2.

   This means that for every value of x there would not always be a value y such that $f(x) = f(y)$, where $x = x$ and $y = f^{-1}(g^{-1}(x))$ meaning it is not injective.

   This would mean that $F2: A \rightarrow A$ would be false implying that F2 is not bijective.

3. F3: $(x,y) \longmapsto (f(x), g(y))$
If f and g are injective then
$f(x) = f(y)$ which implies that $x = y$ and
$g(x) = g(y)$ which implies that $x = y$

   This means that every $y \in B$ has at most one preimage $x \in A$ so $(f(x), g(y))$ would always have at least one value of (x,y) such that $f(x, y) = (f(x), g(y))$, hence proving it surjective:

   $\forall(f(x),g(y)) \in f(A)xg(B)\ \exists(x,y) \in AxB\quad f(x,y) = (f(x),g(y))$

   Because f and g are injective it would mean that every value (for $f(x)$) in $y \in B$ would have at most one preimage in $x \in A$ and (for $g(y)$) every value of $x \in A$ would have at most one preimage in $y \in B$. This would mean that the would only have one value per function and thus proving that every x has a value such that $f(x,y) = f(f(x),g(y))$, thus also proving it injective:

   $f(x,y) = f(f(x),g(y)) \rightarrow (x,y) \in AxB = (f(x),g(y)) \in f(A)xg(B)$ which is $(x,y) = (f(x), g(y))$

   Because F3 is both surjective and injective then F3 is considered bijective.